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Double-loop PI controller design of the DC-DC boost converter with a proposed approach for calculation of the controller parameters

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Abstract

Parameters of digital proportional–integral/proportional–integral–derivative controllers are usually calculated using commonly known conventional methods or solution of discrete-time equations. In literature, a model-based compact form formulation for calculation of discrete-time proportional–integral/proportional–integral–derivative controller parameters has not been come across yet. The proposed model-based compact form formulations are introduced to calculate the proportional–integral parameters in discrete time as a new approach. Generally, different types of control techniques are chosen in similar studies for double-loop control for direct current–direct current boost converter control except proportional–integral/proportional–integral. In this study, double-loop proportional–integral controller is used as a different control method from literature. By this way, the most important advantages of the proposed study are to reduce different design methods to a unique proportional–integral design method and shorten all calculations. The accuracy of the double-loop proportional–integral controller's parameters calculated using the model-based compact form formulations is validated both in simulation and experimental studies under various disturbance effects. Satisfactory performance of the proposed controller under model uncertainty and other cases are comparatively shown with the predefined performance criteria.

Keywords

Compact form formulation, direct current–direct current boost converter, double-loop control, electrical parameter variation, proportional–integral control

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Introduction

Boost-type direct current–direct current (DC-DC) converters are used mostly in industry such as wind energy, solar energy and electrical vehicle systems where output voltage needs to be higher than the input. Controller design of DC-DC boost converters is more complex and difficult than the buck converters because of their non-minimum phase behavior.[1]

proportional–integral/proportional–integral–derivative (PI/PID) control methods are widely used and preferred in most of control applications. PI/PID controller parameters are calculated in literature using heuristic methods such as Ziegler Nichols, particle swarm optimization (PSO), genetic algorithm (GA) and simulated annealing (SA)[2–7] or analytic methods such as frequency response, Bode plot and root-locus technique[8–14] or intelligent methods[15–19] such as neural network (NN) and fuzzy logic.

Boost converter has a Right Half Plane Zero (RHPZ) structure. Dynamic response is very important while designing the controller. However, heuristic methods do not use prior information on dynamic structure of controlled systems thus also may lead to stability problems. Besides, these methods may cause extreme overshoots and sharp rises and falls in the transient and steady-state response of the output.[20]

Although fuzzy logic–based controllers have a good adaptation to non-linear time variant systems, an expert understanding is needed in order to design a reliable

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fuzzy-based controller. Otherwise, a design error may occur. Moreover, a few tools are available for designing the fuzzy-based controllers. Similarly, NN needs well-chosen data and a good network design to get desirable performance from the controller. In addition, intelligent methods are more complex than conventional control methods in terms of implementation in embedded systems.

PI/PID controllers are implemented around the complex conjugate dominant poles ensuring the predefined performance of system response. Stability is also guaranteed with these controllers.

In literature, because of its easy implementation, the PID control is highly desirable in industrial applications likewise DC-DC Converters.[9,21,22] However, single PID control is not enough to ensure the dynamic response of the voltage and current outputs simultaneously. This burden can be eliminated using a double-loop control, which provides the control of both voltage and current outputs.

Generally, different types of cascaded discrete-time control techniques are given in literature such as PI–sliding mode controller (SMC)[1,23] and PI-Fuzzy.[24] Closed-loop analysis and cascaded control of a non-minimum-phase boost converter are applied to regulate the output voltage of a double-loop DC-DC boost converter. Among all these techniques, a double-loop discrete-time PI-PI controller technique has not been studied yet. PI-PI-type controller method has been chosen for this study due to its easy implementation and design structure. Furthermore, the PI parameter calculations of the controllers are getting one step easier with the proposed model-based compact form (MBCF) formulations. Each cascaded controller could be designed with both proposed compact form formulations and conventional design methods. In PI-SMC or PI-Fuzzy–type double-loop controllers, which are used in literature have to be different theorems and design steps for each loop. Therefore, these bring extra design and calculation burden for application engineers and researchers.

Output voltage control of DC-DC Boost converter has two different topologies in literature, one of this topology is entitled by voltage-mode control, and the other one is current-mode control. Although voltage-mode control has a single-loop topology, current-mode control has the double loop, which is the industry standard method of controlling switching power.[25] Therefore, in this study, current-mode control is selected, and the advantages of this control topology could be investigated from industrial applications in literature.[25,26]

There are only a few studies about formulations for PID parameter calculation. Pai et al.[27] present a direct synthesis design (DS-d) formulation for the systems with dead time and inverse response in continuous time. However, these formulations are restricted by a specific type of a process with delay, and the DS-d formulations of PID parameters are only available in continuous time. Previous studies[28–30] present a tuning formula derived especially for PID parameter calculations using phase and gain margins for only continuous time. These formulations are derived for a specific plant and are not generalized for all types of systems.

MBCF formulations have been derived and implemented for a double-loop PI DC-DC boost converter controller design.[31] In the following sections, first, transfer functions of the inner and outer loops are written from small signal analysis. Second, MBCF formulations are obtained. Implementation of the MBCF formulations for the calculation of the PI-PI parameters is also explained. Accuracy of proposed MBCF formulations is validated in three cases with simulation and experimental studies under parameter variations and disturbance effects. The satisfactory responses of the double-loop PI controllers are comparatively shown with the predefined performance criteria.

Model and control of DC-DC boost converter

Proposed double-loop control diagram given in Figure 1 provides instantaneous output voltage and current control simultaneously. Because of its simple and failure tolerant structure and adequate performance in both industrial applications and literature studies,[2,32–36] PI-type controllers are decided to be applied for both loops.

The simplified block diagram of the experimental setup consisting of digital signal processing (DSP)-based double-loop controller together with DC-DC boost converter whose output are connected to feed-resistive loads in parallel is given in Figure 1.

The operating parameters, switching frequency and sampling frequency of the DC-DC boost converter, are adjusted to 100 and 50 kHz, respectively. Insulated Gate Bipolar Transistor (IGBT) current signal i_o is the envelope of the inductor current signal i_L.

Since the proposed MBCF formulations are model based, the model of the system to be controlled must be obtained.

Open-loop transfer functions of the DC-DC boost converter

Considering Figure 1, during the double-loop DC-DC boost converter design, the output voltage and inductor current have to be measured. Therefore, the open-loop transfer functions "$G_1(s)$" and "$G_2(s)$" are derived, respectively, in the following equations.

The transfer function $G_1(s)$ between inductor current and duty ratio "$\tilde{i}_L(s)/\tilde{d}(s)$" is obtained as given in equation (1)

$$G_1(s) = \frac{\tilde{i}_L(s)}{\tilde{d}(s)} = \frac{V_o Cs + 2(1-D)I_L}{LCs^2 + \frac{L}{R}s + (1-D)^2} \quad (1)$$

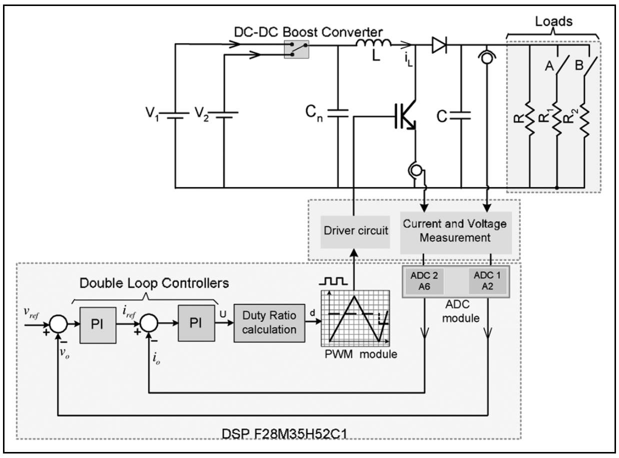

Figure 1. Double-loop control of DC-DC boost converter.

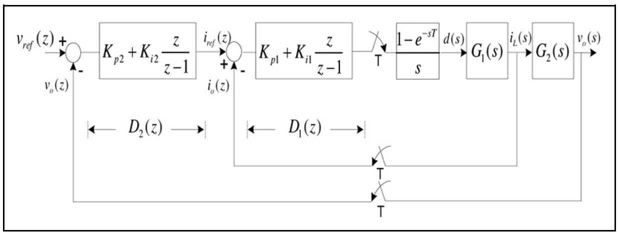

Figure 2. Closed-loop control block diagram of the double-loop control.

Similarly, transfer function between output voltage and the inductor current "$\tilde{v}_o(s)/\tilde{i}_L(s)$" is derived as follows

$$G_2(s) = \frac{\tilde{v}_o(s)}{\tilde{i}_L(s)} = \frac{(1-D)V_o - LI_Ls}{V_oCs + 2(1-D)I_L} \tag{2}$$

The derivation of the inner-loop $G_1(s)$ and outer-loop $G_2(s)$ transfer functions is given in Appendix B.

Double-loop control

Double-loop control shown in Figure 2 using proposed method is explained in two parts; first, proposed compact form formulations are introduced; second, discrete-time PI controllers' parameters have been calculated with the proposed compact form formulations.[1,23,24,37]

When inner and outer-loop controller parameters are calculated, the interaction between two loops must be taken into account. Therefore, in this part, the calculation method of the proposed formulations for the double-loop control system is also explained.

Proposed MBCF formulations. Proposed MBCF formulations using open-loop transfer function are introduced to calculate the PI controller parameters in Figure 3.

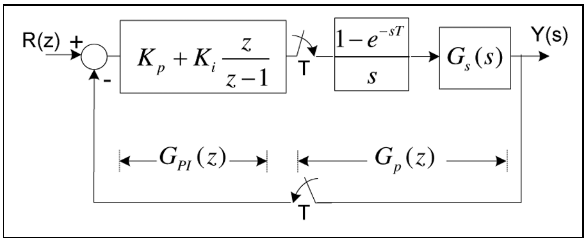

Figure 3. Control block diagram of PI controller system.

The system response $Y(s)$ fulfills the desired transient parameters, overshoot and settling time, when the digital PI controller assigns the two dominant poles of the closed-loop characteristic equation to the predefined poles given in equation (4).

Characteristic equation of closed-loop control system in Figure 3 and redefined poles are given in equations (3) and (4), respectively

$$F(z) = 1 + G_{PI}(z)G_p(z) = 0 \tag{3}$$

$$z_{1,2} = e^{s_{1,2}T} = e^{T(-\xi w_n \pm j w_n \sqrt{1-\xi^2})} = \sigma_{z_{1,2}} + jw_{z_{1,2}} \tag{4}$$

Equation (4) is the definition of s domain and z domain.[38] Where ξ is the damping ratio and w_n is the natural angular frequency.

The control pole "z_1" in equation (4) is replaced by 'z' in characteristic equation (3), and the characteristic equation (3) is arranged as follows

$$F(z_1) = G_{PI}(z_1)G_p(z_1) + 1 = 0 \tag{5}$$

z_1 is written in polar coordinates as given in the following

$$z_1 = \sigma_{z1} + jw_{z1} = |z_1|e^{j\beta} \tag{6}$$

From equation (6), the magnitude and angle of complex value are written as follows

$$|z_1| = \sqrt{\sigma_{z_1}^2 + w_{z_1}^2} \tag{7}$$

$$\beta = \tan^{-1}\left(\frac{w_{z_1}}{\sigma_{z_1}}\right) \tag{8}$$

Similarly, $G_p(z_1)$ complex value can be written in polar coordinates as follows

$$G_p(z_1) = |G_p(z_1)|e^{j\psi} \tag{9}$$

$$\psi = \angle G_p(z_1) \tag{10}$$

If equations (7)–(10) are substituted in equation (5) and rearranged, following equation is obtained

$$K_p + K_i\frac{|z_1|e^{j\beta}}{|z_1|e^{j\beta} - 1} = -\frac{1}{|G_p(z_1)|e^{j\psi}} \tag{11}$$

Proposed MBCF formulations[36] for calculation of the PI parameters are obtained from equation (11) and given in equations (12) and (13), respectively. The derivation of equations (12) and (13) is given in Appendix 1 in detail

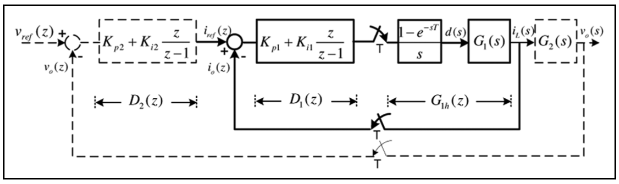

Figure 4. Closed-loop control block diagram of inner loop controller.

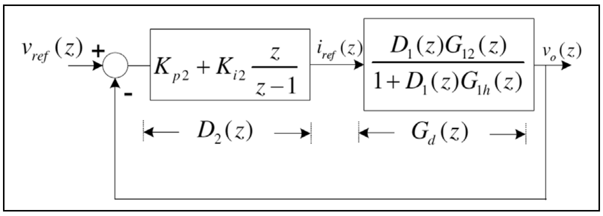

Figure 5. Closed-loop control block diagram of outer-loop controller.

$$K_i = -\frac{\sin\psi}{\left|G_p(z_1)\right|}\frac{|z_1| - 2\cos\beta + \frac{1}{|z_1|}}{\sin\beta} \tag{12}$$

$$K_p = -\frac{\cos\psi}{\left|G_p(z_1)\right|} + \frac{\sin\psi}{\left|G_p(z_1)\right|}\frac{|z_1| - \cos\beta}{\sin\beta} \tag{13}$$

Calculation of discrete-time PI controller parameters. This section guides the designer about how to calculate the PI parameters of inner and outer loops using MBCF formulations.[36]

The PI controller parameters in the loop design are calculated in two steps:

1. The expressions $|z_1|$, β, $\left|G_p(z_1)\right|$ and ψ given in equations (7)–(10) are calculated using defined transient parameter overshoot and settling time.
2. PI parameters are calculated from equations (12) and (13) using the expressions in the first step.

Inner loop controller design. The inner loop controller "$D_1(z)$" is designed due to calculation method mentioned above using $G_1(z)$ and z_1.

First, discrete-time transfer function of $G_{1h}(z)$ is obtained using the DC-DC boost converter parameters and given in equation (14). These parameters are used in both simulation and experimental studies. Inductor value $L = 620\,\mu\text{F}$, capacitor value $C = 1640\,\mu\text{F}$, duty cycle $D = 0.33$, desired output voltage $V_o = 18\,\text{V}$, input voltage $V_i = 12.2\,\text{V}$, steady-state inductor current $I_L = 1.289\,\text{A}$, load at operating point $R = 20.6\,\Omega$ and sampling time $T = 20\,\mu\text{s}$

$$\begin{aligned} G_{1h}(z) &= Z\left\{\frac{1-e^{-sT}}{s}\frac{V_0Cs + 2(1-D)I_L}{LCs^2 + \frac{L}{R}s + (1-D)^2}\right\} \\ &= \frac{0.5808z - 0.5801}{z^2 - 1.99z + 0.994} \end{aligned} \tag{14}$$

Second, dominant pole z_1 is calculated using performance criteria; settling time $t_{s1} = 4\,\text{ms}$ and the percentage overshoot $OS = 4.33\%$ and sampling time $T = 20\,\mu\text{s}$. Inductor current is controlled in the inner loop. Since the reference current I_{ref} is produced by the outer loop, the inner-loop response must be faster than the outer-loop response. Settling time, which is a performance parameter, should be selected by considering this restriction.

The complex conjugate dominant poles are calculated from equation (4) using defined ξ and w_n values

$$z_{1,2} = 0.9731 \pm 0.0262j \tag{15}$$

The values of expressions in equations (7)–(10) are calculated as follows

$$\begin{aligned} |z_1| &= 0.9735 \\ \beta &= 0.0269 \\ |G_{1h}(z_1)| &= 15.2049 \\ \psi &= -2.5065 \end{aligned} \tag{16}$$

The K_p and K_i parameters of $D_1(z)$ controller in the inner loop are obtained using the MBCF formulations (12) and (13). The results are as follows

$$\begin{aligned} K_{p1} &= 0.0909 \\ K_{i1} &= 0.0021 \end{aligned} \tag{17}$$

Outer-loop controller design. Design of the outer-loop controller $D_2(z)$ according to calculation method mentioned above is slightly different than the inner loop design. First, designer has to be calculated $G_d(z)$ transfer function which is given in Figure 5.

$G_{12}(z)$ transfer function is calculated as

$$\begin{aligned} G_{12}(z) &= Z\left\{\frac{1-e^{-sT}}{s}(G_1(s)G_2(s))\right\} = \\ &\frac{-0.01286z^2 + 0.3142z - 0.01855}{z^3 - 2.998z^2 + 2.996z - 0.9982} \end{aligned} \tag{18}$$

It can be seen that the order of the $G_d(z)$ is equal to seven when Table 1 and $G_d(z)$ transfer function are taken into account

$$G_d(z) = \frac{v_o(z)}{i_{ref}(z)} = \frac{D_1(z)G_{12}(z)}{1 + D_1(z)G_{1h}(z)}$$

the discrete-time closed-loop transfer function of the outer loop is given as follows

$$\frac{v_o(z)}{i_{ref}(z)} = \frac{-0.0012z^6 + 0.0078z^5 - 0.020z^4 + 0.028z^3 - 0.022z^2 + 0.009z - 0.001}{z^7 - 6.94z^6 + 20.6z^5 - 34.1z^4 + 33.8z^3 - 20.1z^2 + 6.6z - 0.94} \tag{19}$$

Table 1. Order of the transfer functions.

Transfer function	Order
$G_{12}(z)$	3
$G_{1h}(z)$	2
$D_1(z)$	1
$G_d(z)$	7

The performance criteria of the outer loop are determined as settling time $t_{s2} = 16.2\,\text{ms}$, percentage overshoot $OS = 4.33\%$ and sampling time $T = 20\,\mu\text{s}$. In this study, $t_{s2} = 4.05t_{s1}$ has been chosen.

The values of expressions in equations (7)–(10) are calculated using the defined performance criteria.

The K_p and K_i parameters of the outer-loop $D_2(z)$ controller are obtained using the MBCF formulations (12) and (13). The results are given as follows

$$\begin{aligned} K_{p2} &= 1.2863 \\ K_{i2} &= 0.0134 \end{aligned} \tag{20}$$

Experimental and simulation results of the open-loop transfer function verification and the transient and steady-state analysis of the output voltage/current are given in the following section. In the experimental studies, since IGBT current signal i_o is the envelope of the inductor current signal i_L, instantaneous value i_o is measured instead of instantaneous value i_L.

Simulation and experimental results

To demonstrate the performance of the proposed MBCF formulations, switching frequency and sampling frequency of 50kHz and 100kHz, respectively, are chosen. The DC-DC boost converter is implemented using the nominal values: 18 V, 2 A, $L = 624\,\mu\text{H}$, $C = 1640\,\mu\text{F}$, Diode 30CTQ060STRLPBFCT and IGBT IRFR3607TRPBFCT DPAK. Measurement and control algorithms are executed by 80 MHz DSP F28M35H52C1–based development kit. Experimental setup is given in Figure 6.

Experimental and simulation studies have three stages:

1. Model verification.
2. Performance analysis of the proposed method.
3. Performance in electrical parameter variations.

Model verification

Model verification of the open-loop transfer function of $\tilde{v}_o(s)/\tilde{d}(s)$ shown in Figure 2 is performed under a defined duty ratio operating point using the given electrical parameters. Simulation and experimental results, which validate the model accuracy of the DC-DC boost converter, are given in Figure 7.

The proposed method is model based. Hence, the error should be lower than the acceptable value.

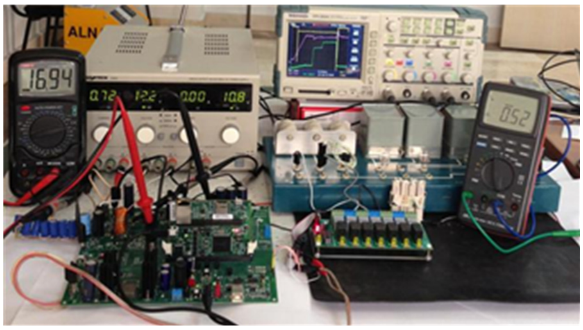

Figure 6. Experimental setup.

Table 2 shows that the simulation and experimental result errors are lower than 1%. This shows that the model used for controller design is appropriate. Moreover, the robustness of the DC-DC boost converter against electrical parameter variations is given in case 3.

Performance analysis of the proposed method

DC-DC boost output voltage has been regulated via PI controllers whose coefficients are calculated using the proposed method for inner and outer loops. Because of its analytical calculation methodology, which is based on pole placement, MBCF formulations are also ensured of the stability of the control system. Further details could be found in Ozdemir and Erdem.[31] The performance analysis is given in detail with both simulation and experimental results for the three cases as given below.

Case 1: dynamic response performance. In this section, dynamic responses of the DC-DC boost converter's output voltage are analyzed as follows:

1. Step changes in the reference input.
2. The input voltage sag during step changes in the reference input and step load changes for both cases.

Step changes in the reference input. While DC-DC boost converter is operated at the 14 V steady-state value, the input voltage is step changed to 18 V with load change in $t = 2\,\text{ms}$. Related waveform for the responses is depicted in Figure 8.

The values related to transient and steady-state responses are obtained by considering the simulation and the experimental waveforms shown in Figure 8. These results are tabulated and comparatively given in Table 3.

Two different assessments can be made for the values given in Table 3:

1. Simulation results are validated by experimental results.

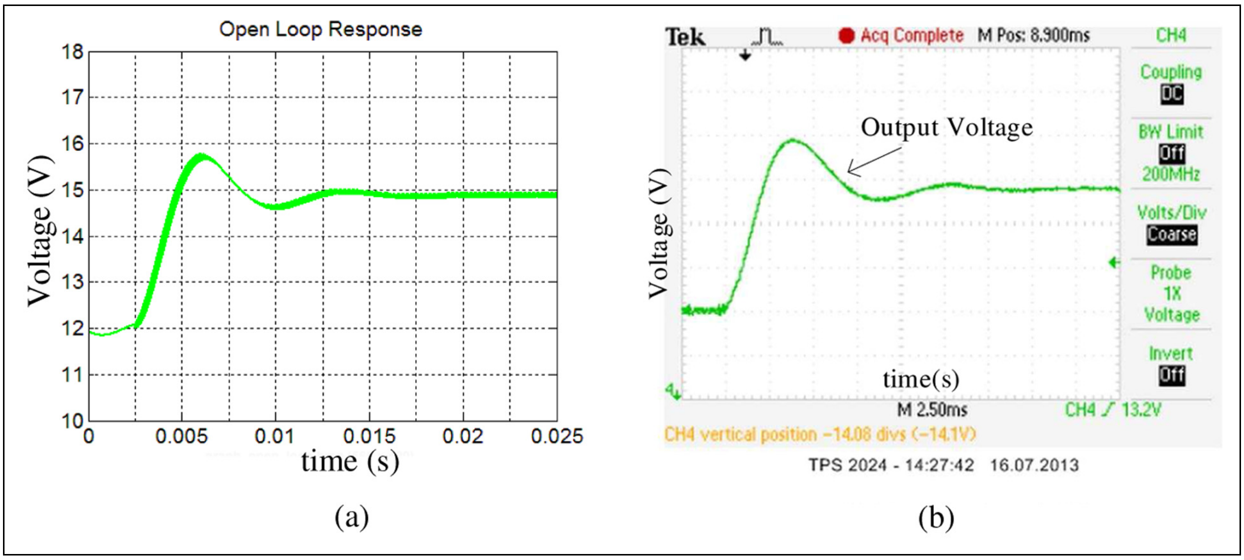

Figure 7. Open-loop transfer function response: (a) simulation and (b) experimental results (12.3 V input voltage, 20.6 Ω load and 0.2 duty cycle rate).

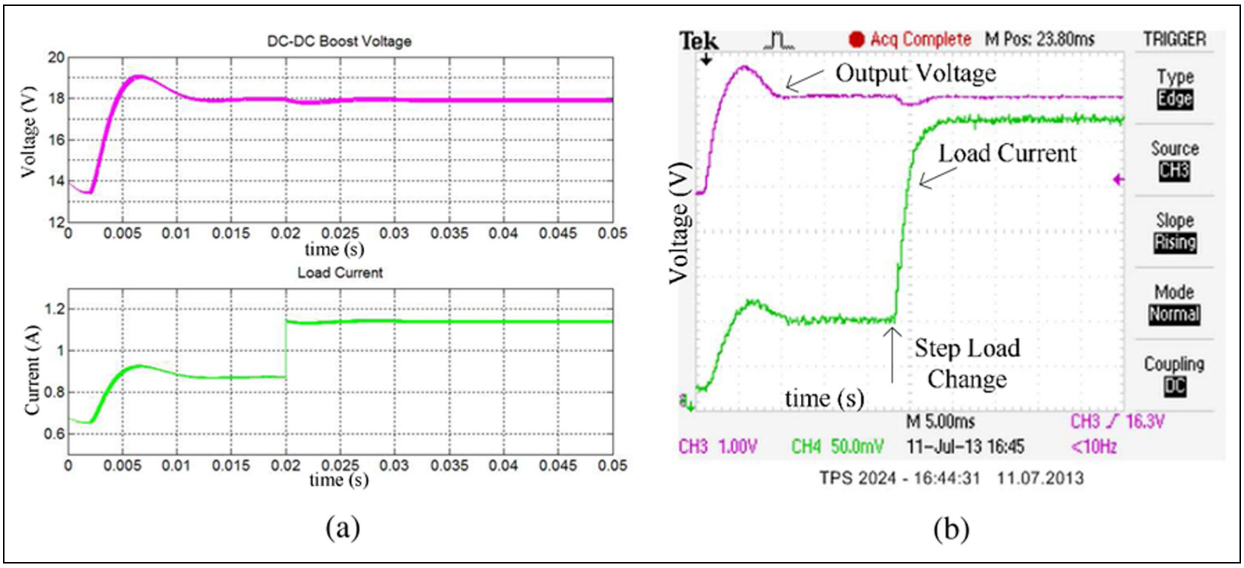

Figure 8. Step changes in the reference input (22%) and load (25%): (a) simulated output voltage and current responses and (b) experimental output voltage and current responses (input voltage is stepped up to 14–18 V at $t = 2$ ms and the load is increased by 25% at $t = 20$ ms).

2. Both simulation and experimental results ensured the following predefined performance criteria.

Settling time $t_{sper} > t_{sreal}, t_{ssim}$;
Percentage overshoot $OS_{per} > OS_{real}, OS_{sim}$; peak voltage value $V_{opper} > V_{opreal}, V_{opsim}$;
Desired output voltage $V_{oper} = V_{osim} = V_{oreal}$.

Input voltage sag. Simulation and experimental resultant waveforms for the voltage sag, output voltage and the load current are shown together in Figure 9(a) and (b), respectively.

The input voltage sag in Figure 9(a) is generated by signal builder block in MATLAB/Simulink. The step loads are switched on and off between time intervals of 40 and 74 ms. Steady-state values of output voltage are same (16 V), and percentage overshoots are measured as 4.3% and 3.75% for simulation and experimental study, respectively. It can be clearly seen in Figure 9 that the disturbance effect of voltage sag is suppressed by the controller, and thus, the system response does not deteriorate.

Case 2: soft start response. Soft start provides wide scale in the DC-DC boost converter reference input voltage change. In addition, soft start avoids over current and protect power switching component. In order to demonstrate the performance of the proposed method,

Table 2. Transient parameter comparison of the open-loop response.

Parameters	Simulation	Experimental	Error (%)
Peak value	$V_{opsim} = 15.8\,V$	$V_{opreal} = 15.9\,V$	0.62
Settling time (2%)	$t_{ssim} = 7.5\,ms$	$t_{sreal} = 7.44\,ms$	0.81
Percentage overshoot	$OS_{sim} = 6.04\%$	$OS_{real} = 6.71\%$	0.67
Steady-state value	$V_{sssim} = 14.9\,V$	$V_{ssreal} = 14.9\,V$	0

Table 3. Transient response variables under reference and load variation.

Parameters	Simulation	Experimental	Performance criteria
Peak value	$V_{opsim} = 18.5\,V$	$V_{opreal} = 18.7\,V$	$V_{opper} = 18.77\,V$
Settling time	$t_{ssim} = 9\,ms$	$t_{sreal} = 7\,ms$	$\mathbf{t_{sper}} = 16\,ms$
Steady-state value	$V_{osim} = 18\,V$	$V_{oreal} = 18\,V$	$V_{oper} = 18\,V$
Percentage Overshoot	$OS_{sim} = 2.78\%$	$OS_{real} = 3.89\%$	$OS_{per} = 4.33\%$

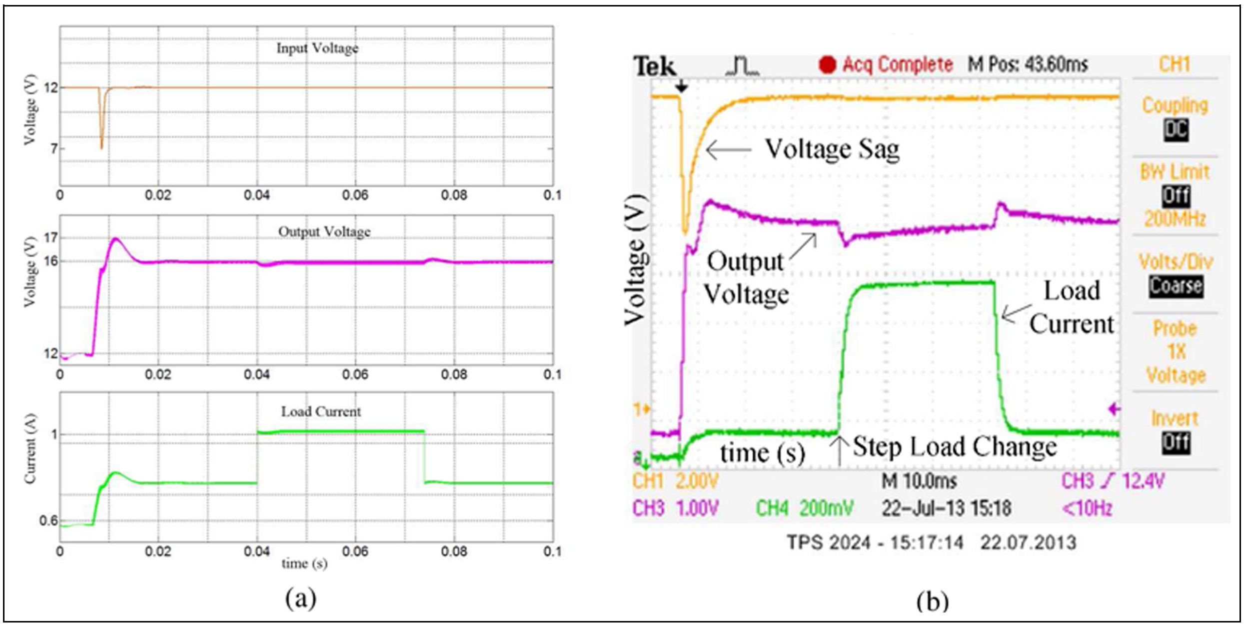

Figure 9. Input voltage sag and output voltage and load step change: (a) simulation and (b) experimental results—reference voltage step changes from 12 to 16 V at 8 ms, load step change increased 25% (20–15 Ω) and reduced 25%, and voltage sag of 58% (12–6.96 V).

two different disturbances are sequentially applied within soft start period. In experimental application, soft start is achieved with DSP-based software, and no additional hardwares are used. Transient and steady-state responses are shown in Figure 10.

The parameter values given in Table 4 and Figure 10 shows that the controller performance under disturbance effects assures the desired performance in both simulation and experimental studies.

The transient parameters such as settling time, steady-state value of output voltage, overshoot and undershoot illustrate that the system controlled by proposed method has a satisfying dynamic response.

Table 4. Soft start results for both simulation and experimental.

Parameters	Simulation	Experimental
Settling time	$t_{ssim} = 7.2\,ms$	$t_{sreal} = 7\,ms$
Steady-state value	$V_{osim} = 19\,V$	$V_{oreal} = 19\,V$

Case 3: Robustness against the electrical parameter variations. In experimental applications, generally those parameters (L and C) cannot be measured accurately, or their nominal values may be changed by environmental effects. Proposed MBCF formulations are

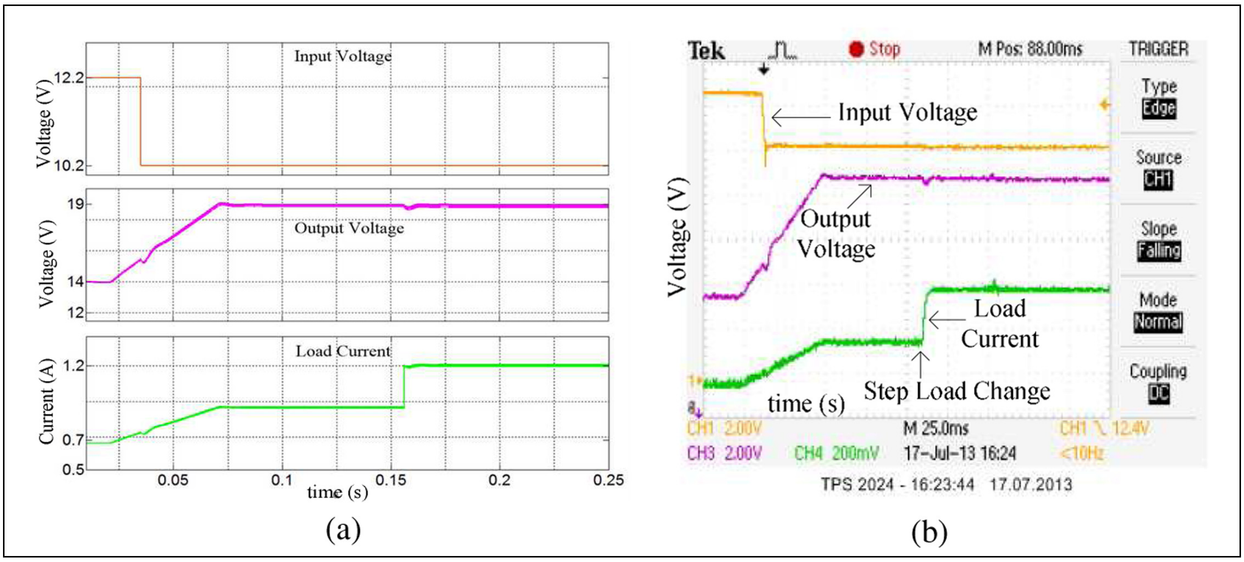

Figure 10. Soft start response while input voltage is decreased (16%) and load step is changed (25%): (a) simulation results and (b) experimental results—output voltage reference changes from 14 V to 19 V; soft start duration is 50 ms; source voltage changes from 12.2 to 10.2 V (decreased by 16%).

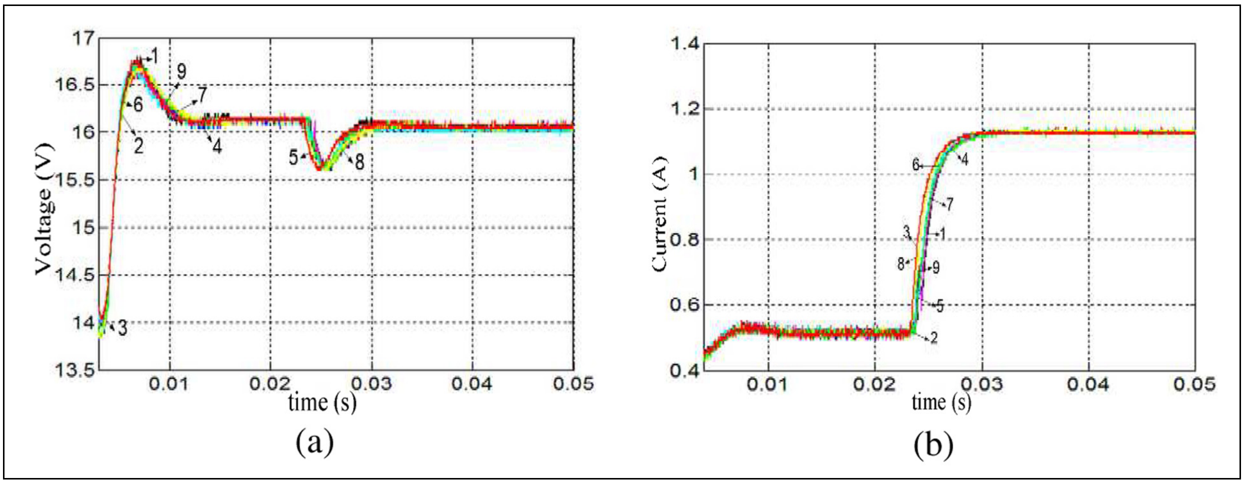

Figure 11. Experimental transient response of the DC-DC boost converter output voltage and current under ±10% electrical parameter change: (a) voltage output and (b) load current.

model based. In this case, the PI controller coefficients are calculated using MBCF formulations for nominal + 10% and −10% electrical parameter L and C variation combinations. Each calculated PI controller coefficients are applied to the DC-DC boost converter in real time and the results are compared. Comparison results show that the MBCF formulations are robust against the variation in the electrical parameter values. The L (inductor) and C (Capacitor) parameters on the DC-DC boost converter circuit shown in Figure 1 are not changed during this case. Output voltage and load current waveforms are obtained as shown in Figure 11 for the step changes in load and reference input voltage.

Figure 11(a) waveform shows that the DC-DC boost converter's steady-state value of the output voltage remains within ±1% tolerance dependent on the parameter variation. This tolerance is within the ±2% band which is predefined for calculating the settling time value t_s. As an example, nominal values of the L (inductor) and C (capacitor) are $L = 620\,\mu H$ and $C = 1640\,\mu F$, respectively. Waveform number 1 shows the DC-DC boost converter output voltage, and current results with nominal value of the electrical parameters and the PI parameters of the double-loop controller are $K_{p1} = 0.0909$ and $K_{i1} = 0.0021$ for the inner loop and $K_{p2} = 1.2868$ and $K_{i2} = 0.0134$ for the outer loop. However, to demonstrate the environmental effect on the parameter variation, the electrical parameters are assumed to be $L = 558\,\mu H$ and $C = 1476\,\mu F$ while calculating the PI controller parameters using MBCF formulations. The PI coefficients

Table 5. Benchmark table.

Study	Method	System type	Mode	Settling time–output voltage	Overshoot–output voltage	Settling time–output current	Overshoot–output current	Input voltage step change	Output voltage overshoot or undershoot via input voltage step change	S – R
Chen[1]	PID-Sliding Mode	DC/DC Buck-Boost Converter	CCM	0.025	13%	0.0025	0	20%	10%(o)	S + R
Salimi et al.[39]	VSL + PI–ABC	DC/DC Buck-Boost Converter	DCM	0.023	12%	0.023	12%	30%	8%(o)	S + R
Gundemir[40]	PI + Sliding Mode Control	Boost Converter	CCM	0.018	10.7%	0.018	10%	10%	17%(u)	S
Salimi[41]	PID + PID	Flyback Controller	CCM	0.8	15%(u)	0.8	35%	50%	15%(u)	S
Ren et al.[37]	PI + switching base CLF	Boost Converter	CCM	0.2	1.33%	NA	NA	NA	6.67%(u)	S + R
Agorreta et al.[24]	PI + Fuzzy	Boost Converter	CCM + DCM	0.15	50%	0.15	50%	NA	NA	S + R
Proposed study	PI + PI	Boost Converter	CCM	0.009	2.78%	0.009	2.78%	16%	3%(o)	S + R

NA: not available; CCM: continuous conduction mode; DCM: discontinuous conduction mode; VSL: variable structure linear controller; ABC: adaptive back stepping current; S: simulation; R: real time; (o): overshoot; (u): undershoot; CLF: common Lyapunov function.

are recalculated for the inner loop as $K_{p1} = 0.1142$ and $K_{i1} = 0.0033$ and for the outer loop as $K_{p2} = 1.3075$ and $K_{i2} = 0.0171$. DC-DC boost converter output voltage and current waveform are shown with number 9 in Figure 11(a) and (b). Despite the change in the double-loop controller parameters, Figure 11 shows that system response is robust against variation in the electrical parameter values or incorrect measurements and environmental changes. Case 3 results also indicate that the $\pm 10\%$ parameter variation does not affect the controller performance, nevertheless the MBCF formulations are model based.

Benchmark with previous studies. A benchmark table is created with similar previous studies to compare proposed study. Six different studies selected from literature and results are given in Table 5.

Conclusion

This study is presented to indicate that the double-loop PI controller's parameters could be accurately calculated using the proposed MBCF formulations. Case 1–3 studies validate that the double-loop PI controllers fulfill the desired performance criteria despite step changes in reference input and disturbances such as voltage sag, input voltage step change, load step change and soft start.

In this study, the simplest model of the DC-DC boost converter is obtained with linearization and using the ideal L and C equivalent circuits. Although the PI controller's parameters are calculated by proposed MBCF formulations using the simplest DC-DC boost converter model, experimental results are approved by the simulation results. Furthermore, the robustness of the MBCF formulations against the variations such as electrical parameters (L and C), capacitor equivalent series resistance (ESR), non-linearity of pulse width modulation (PWM) modulator, quantization effects and so on are demonstrated by regulating DC-DC boost converter's output voltage within $\pm 1\%$ tolerance as shown in Figure 11.

Simulation results for all cases mentioned above were experimentally validated with predefined performance criteria. Proposed MBCF formulations provide easy and accurate calculation of PI controller parameters for industrial application engineers and researchers.

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References

1. Chen Z. Double loop control of buck-boost converters for wide range of load resistance and reference voltage. *IET Control Theory A* 2012; 6(7): 90.
2. Sanjeevikumar P and Rajambal K. Extra high voltage DC-DC boost converters topology with simple control strategy. *Model Simul Eng* 2008; 2008: 593042.
3. Hägglund T and Åström KJ. Revisiting the Ziegler-Nichols tuning rules for PI control. Part II: the frequency response method. *Asian J Control* 2008; 6(4): 469–482.
4. Kalaviani R and Lakshami P. Biogeography-based optimization of PID tuning parameters for the vibration control of active suspension system. *J Control Eng Appl Inform* 2014; 16(1): 31–39.
5. Nasri M, Nezamabadi-pour H and Maghfoori M. A PSO-based optimum design of PID controller for a linear brushless DC motor. *J World Acad Sci Eng Technol* 2007; 2: 211–215.
6. De Moura Oliveira PB. Modern heuristics review for PID control systems optimization: a teaching experiment. In: *2005 international conference on control and automation*, Budapest, 26–29 June 2005, pp.828–833. New York: IEEE.
7. Yachen Z and Yueming H. On PID controllers based on simulated annealing algorithm. In: *2008 27th Chinese control conference*, Kunming, China, 16–18 July 2008, pp.225–228. New York: IEEE.
8. Bryant B and Kazimierczuk MK. Open-loop power-stage transfer functions relevant to current-mode control of boost PWM converter operating in CCM. *IEEE T Circuits I* 2005; 52(10): 2158–2164.
9. Guo L, Hung JY and Nelms RM. Evaluation of DSP-based PID and fuzzy controllers for DC–DC converters. *IEEE T Ind Electron* 2009; 56(6): 2237–2248.
10. Shen Z, Yan N and Min H. A multimode digitally controlled boost converter with PID autotuning and constant frequency/constant off-time hybrid PWM control. *IEEE T Power Electr* 2011; 26(9): 2588–2598.
11. Maksimovic D and Zane R. Small-signal discrete-time modeling of digitally controlled PWM converters. *IEEE T Power Electr* 2007; 22: 6: 2552–2556.
12. Guo L, Hung JY and Nelms RM. Comparative evaluation of sliding mode fuzzy controller and PID controller for a boost converter. *Electr Pow Syst Res* 2011; 81(1): 99–106.
13. Hua C, Lin J and Shen C. Implementation of a DSP-controlled photovoltaic system with peak power tracking. *IEEE T Ind Electron* 1998; 45(1): 99–107.
14. Trejos A, Gonzalez D and Ramos-Paja CA. Modeling of step-up grid-connected photovoltaic systems for control purposes. *J Energ* 2012; 5(12): 1900–1926.
15. Seo KW and Choi H-H. Simple fuzzy PID controllers for DC-DC converters. *J Electr Eng Technol* 2012; 7(5): 724–729.
16. Volosencu C, Danica R and Pasgianos GD. *Introduction to PID controllers: theory, tuning and application to frontier areas*. (ed. Panda RC). London: InTech, 2012.
17. Luoren L and Jinling L. Research of PID control algorithm based on neural network. *Enrgy Proced* 2011; 13: 6988–6993.
18. Agee JT, Davidson IE and Kombani LT. Intelligent proportional integral control of a polar axis solar tracker. In: *2016 Clemson University power systems conference*, Clemson, SC, 8–11 March 2016, pp.1–6. New York: IEEE.
19. Ibrahim IN and Al Akkad MA. Exploiting an intelligent fuzzy-PID system in nonlinear aircraft pitch control. In: *2016 international Siberian conference on control and communications, Moscow*, Russia, 12–14 May 2016, pp.1–7. New York: IEEE.
20. Stefanutti W, Mattavelli P, Saggini S, et al. A PID autotuning method for digitally controlled DC-DC boost converters. *In: European conference on power electronics and applications*, Dresden, 11–14 September 2005, pp.10–17. New York: IEEE.
21. Guo L. Implementation of digital PID controllers for DC-DC converters using digital signal processors. In: *IEEE international conference on electro/information technology*, Chicago, IL, 17–20 May 2007, pp.306–311. New York: IEEE.
22. Arulselvi S, Uma G and Chimdambaram M. Design of PID controller for boost converter with RHS zero. In: *4th international power electronics and motion control conference, Xi'an, China*, 14–16 August 2004, pp.532–537. New York: IEEE.
23. Chen Z, Gao W, Hu J, et al. Closed-loop analysis and cascade control of a nonminimum phase boost converter. *IEEE T Power Electr* 2011; 26(4): 1237–1252.
24. Agorreta JL, Reinaldos L, Gonzalez R, et al. Fuzzy switching technique applied to PWM boost converter operating in mixed conduction mode for PV systems. *IEEE T Ind Electron* 2009; 56(11): 4363–4373.
25. Lee SW. *Practical feedback loop analysis for current-mode boost converter*. Application Report, March 2014. Dallas, TX: Texas Instruments.
26. Hegarty T. Current-mode control stability analysis for DC-DC converters (part 1). *How2Power Today*. June 2014; 1–7.
27. Moradi MH. New techniques for PID controller design. *In: IEEE conference on control applications*, Istanbul, 25–25 June 2003. New York: IEEE.
28. Pai N-S, Chang S-C and Huang C-T. Tuning PI/PID controllers for integrating processes with deadtime and inverse response by simple calculations. *J Process Contr* 2010; 20(6): 726–733.
29. Ho WK, Hang CC and Zhou J. Self-tuning PID control of a plant with under-damped response with specifications on gain and phase margins. *IEEE T Contr Syst T* 1997; 5(4): 446–452.
30. Ho W, Hang C and Cao L. Tuning of PID controllers based on gain and phase margin specifications. *Automatica* 1995; 31(3): 3–8.
31. Ozdemir A and Erdem Z. A new approach for calculation of PID parameters with model based compact form formulations. *Elektronika Ir Elektrotechnika* 2014; 20(3): 3–10.
32. Ntogramatzidis L and Ferrante A. Exact tuning of PID controllers in control feedback design. *IET Control Theory Appl* 2011; 5(4): 565.
33. Eltamaly AM, El-Tamaly HH and Enjeti P. An improved maximum power point tracker for photovoltaic energy systems. In: *The 2nd Minia international conference for advanced trends in engineering*, El-minia, 7–9 April 2002. Egypt: Elminia University.
34. Hua C and Chihming S. Study of maximum power tracking techniques and control of DC/DC converters for photovoltaic power system. In: *29th annual IEEE power*

electronics specialists conference, Fukuoka, Japan, 22–22 May 1998. New York: *IEEE*.
35. Koutroulis E, Kalaitzakis K and Voulgaris NC. Development of a microcontroller-based, photovoltaic maximum power point tracking control system. *IEEE T Power Electr* 2001; 16(1): 46–54.
36. Mutoh N, Masahiro O and Takayoshi I. A method for MPPT control while searching for parameters corresponding to weather conditions for PV generation systems. *IEEE T Ind Electron* 2006; 4(53): 1055–1065.
37. Ren H, Xin G and Zi Y. Double loop control of boost converter based current switching controller and voltage compensator. In: *International conference on electronics, computers and artificial intelligence*, Bucharest, 25–27 June 2015. New York: IEEE.
38. Ogata K. *Discrete time control systems*. 2nd ed. Upper Saddle River, NJ: Prentice-Hall, 1995, pp.205.
39. Salimi M, Soltani J, Zakipour A, et al. Two-loop adaptive and nonlinear control of the DC-DC boost converter in discontinuous conduction mode. In: *4th power electronics, drive systems and technologies conference*, Tehran, 13–14 February 2013, pp.164–169. New York: IEEE.
40. Gundemir H. Sliding mode control of DC-DC boost converter. *J Appl Sci* 2005; 5(3): 588–592.
41. Salimi M. Two-loop control of the DC-DC flyback converter. In: *3rd international conference on advancements in electronics and power engineering*, Kuala Lumpur, Malaysia, 8–9 January 2013, pp.230–232. Planetary Scientific Research Centre (PSRC).
42. Smitha K, Aryar PG, Bijlwan R, et al. Steady state analysis of PID controlled boost converter using state space averaging technique. (Special issue published by Multidisciplinary Journal of Research in Engineering and Technology) In: *National Conference*, 2015, pp.100–110. Pune, India: Green Publications.

Appendix 1

Where Euler's formula expressions are substituted into equation (11) and rearranged as follows

$$K_p + K_i \frac{|z_1|(\cos\beta + j\sin\beta)}{|z_1|(\cos\beta + j\sin\beta) - 1} = \frac{-\cos\psi + j\sin\psi}{|G_p(z_1)|} \tag{21}$$

where K_p and K_i parameters are obtained by rearranging them as the real part and imaginary part as given in equations (22) and (23), respectively

Real part

$$K_p = -K_i \frac{|z_1|^2 - |z_1|\cos\beta}{|z_1|^2 - 2|z_1|\cos\beta + 1} - \frac{\cos\psi}{|G_p(z_1)|} \tag{22}$$

Imaginary part

$$K_i \frac{-|z_1|\sin\beta}{|z_1|^2 - 2|z_1|\cos\beta + 1} = \frac{\sin\psi}{|G_p(z_1)|} \tag{23}$$

where K_i parameter is obtained as follows

$$K_i = -\frac{\sin\psi}{|G_p(z_1)|} \frac{|z_1|^2 - 2|z_1|\cos\beta + 1}{-|z_1|\sin\beta} \tag{24}$$

when the K_i parameter is substituted in equation (22), K_p is obtained as given in equation (25)

$$K_p = -\frac{\cos\psi}{|G_p(z_1)|} + \frac{\sin\psi}{|G_p(z_1)|} \frac{|z_1| - \cos\beta}{\sin\beta} \tag{25}$$

Further information about MBCF formulations could be found in Ozdemir and Erdem.[31]

Appendix 2

Derivation of inner- and outer-loop transfer functions from state equations[42] is given below

$$\begin{bmatrix} \frac{d\tilde{i}_L(t)}{dt} \\ \frac{d\tilde{v}_o(t)}{dt} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \frac{-(1-D)}{L} \\ \frac{1-D}{C} & -\frac{1}{RC} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \tilde{i}_L \\ \tilde{v}_o \end{bmatrix}}_{x} + \underbrace{\overbrace{\begin{bmatrix} \frac{V_o}{L} \\ -\frac{I_L}{C} \end{bmatrix}}^{B_1} \tilde{d} + \overbrace{\begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}}^{B_2} \tilde{v}_g}_{B.u} \tag{26}$$

$$\tilde{i}_L = \underbrace{[1 \quad 0]}_{C_1} \begin{bmatrix} \tilde{i}_L \\ \tilde{v}_o \end{bmatrix} \tag{27}$$

$$\tilde{v}_0 = \underbrace{[0 \quad 1]}_{C_2} \begin{bmatrix} \tilde{i}_L \\ \tilde{v}_o \end{bmatrix} \tag{28}$$

From the state equations (26), (27) and (28), $G_n(s) = C_i(sI - A)^{-1}B_1$ ($n = 0, 1, 2$ and $i = 1, 2$) is used to obtain $G_1(s)$ and $G_0(s) = \tilde{v}_o(s)/\tilde{d}(s)$ open-loop transfer functions

$$G_1(s) = \frac{\tilde{i}_L(s)}{\tilde{d}(s)} = C_1(sI - A)^{-1}B_1 \tag{29}$$

State equation coefficients C_1, A and B_1 are replaced in equation (29) and can be rewritten as

$$G_1(s) = [1 \quad 0]\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & \frac{-(1-D)}{L} \\ \frac{(1-D)}{C} & -\frac{1}{RC} \end{bmatrix}\right)^{-1} \begin{bmatrix} \frac{V_o}{L} \\ -\frac{I_L}{C} \end{bmatrix} \tag{30}$$

Rewriting equation (30), $G_1(s)$ is obtained as follows

$$G_1(s) = \frac{\tilde{i}_L(s)}{\tilde{d}(s)} = G_1(s) = \frac{V_o Cs + 2(1-D)I_L}{LCs^2 + \frac{L}{R}s + (1-D)^2} \tag{31}$$

Where small derivations in $\tilde{v}_0$ is assumed as zero

$$\tilde{v}_0 = 0 \rightarrow \frac{V_0}{R} = (1-D)I_L \tag{32}$$

Outer-loop transfer function $G_2(s) = \tilde{v}_o(s)/\tilde{i}_L(s)$ is obtained from

$$G_2(s) = \frac{G_0(s)}{G_1(s)} = \frac{\frac{\tilde{v}_o(s)}{\tilde{d}(s)}}{\frac{\tilde{i}_L(s)}{\tilde{d}(s)}} = \frac{\tilde{v}_o(s)}{\tilde{i}_L(s)} \tag{33}$$

$$G_0(s) = \frac{\tilde{v}_o(s)}{\tilde{d}(s)} = C_2(sI - A)^{-1}B_1 \tag{34}$$

For the derivation of $G_0(s)$ state equation, coefficients C_2, A and B_1 are replaced in equation (34) and rewrite

$$G_0(s) = \begin{bmatrix} 0 & 1 \end{bmatrix} \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & \frac{-(1-D)}{L} \\ \frac{(1-D)}{C} & -\frac{1}{RC} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{Vo}{L} \\ -\frac{I_L}{C} \end{bmatrix} \tag{35}$$

$G_0(s)$ is derived from equation (35) and obtained as follows

$$G_0(s) = \frac{\tilde{v}_o(s)}{\tilde{d}(s)} = \frac{(1-D)V_o - LI_L s}{LCs^2 + \frac{L}{R}s + (1-D)^2} \tag{36}$$

$G_2(s)$ outer-loop transfer function is derived from equation (33) and obtained as follows

$$G_2(s) = \frac{G_0(s)}{G_1(s)} = \frac{\tilde{v}_o(s)}{\tilde{i}_L(s)} = \frac{(1-D)V_o - LI_L s}{V_o Cs + 2(1-D)I_L} \tag{37}$$